

13 Factorials Based Best Questions

$$5! = 120$$

$$5! = 5 \times 4 \times 3 \times 2 \times 1$$

$$n! = ?$$

$$n! = (n) \times (n-1) \times (n-2) \times (n-3) \dots$$

$$= (n-1+1) \times (n-2+1) \times (n-3+1) \dots \times (n-n+1)$$

In General $(n-r+1)$
 $\downarrow$
variable

$$n! = n \times (n-1)!$$

$$(n-1)! = (n-1) \times (n-2)!$$

* write 1 to 10!

$$1! = 1$$

$$2! = 2$$

$$3! = 6$$

$$4! = 24$$

$$5! = 120$$

$$6! = 120 \times 6 = 720$$

$$7! = 720 \times 7 = 5040$$

$$8! = 5040 \times 8 = 40320$$

$$9! = 362880$$

$$10! = 3628800$$

* $4!$ & onwards every factorial is $\neq 4$
 & $\neq 0$

* $5!$ & onwards every factorial as its
 unit digit $= 0$

3 Questions :-

① If $a! + b! + c! + d! + e!$ is a
 two digit number, find the maximum value

Solⁿ :- put

$$a! = 1! = 1$$

$$b! = 2! = 2$$

$$c! = 3! = 6$$

$$d! = 4! = 24 \quad \checkmark \quad \text{it is max 2 digit}$$

$$e! = 5! = 120 \quad \times \quad \therefore \text{not 2 digit}$$

Ans :- 4

② If the unit digit of $a! + b! + c!$ is 9,
 then find the value of $(a!)(b!)(c!)$.

Solⁿ Given

$$a! + b! + c! = 9$$

$$1! + 2! + 3! = 9$$

$$1 + 2 + 6 = 9$$

$$\therefore (a!)(b!)(c!) = ?$$

$$(1)(2)(6) = 12$$

Ans = 12.

③ What will be the ten's digit of
 $1! + 2! + 3! + 4! + 5! + \dots + 10!$?

Solⁿ

$$1! = 01$$

$$2! = 02$$

$$3! = 06$$

$$4! = 24$$

$$5! = 20 \rightarrow \text{write only ten's digit}$$

$$6! = 20$$

means last
2 digits.

$$7! = 40$$

$$8! = 20$$

$$9! = 80$$

$$10! = 00$$

$$\underline{213}$$

Ans is 1 this is last 2nd digit (ten's)

④ Find the unit digit of

$$1! + 2! + 3! + 4! + \dots + 50!$$

Solⁿ

w.k.T

$$1! = 1$$

$$2! = 2$$

$$3! = 6$$

$$4! = 24$$

$$5! = 120$$

$$6! = 0$$

$$7! = 0$$

$$8! = 0$$

$$9! = 0$$

$$10! = 0$$

$$\underline{13}$$

w.k.T 5! onwards
every unit digit
are 0.

Ans 3

⑤ Find the unit digit of.

$$(1!)^1 + (2!)^2 + (3!)^3 + \dots + (89!)^{89}$$

Solⁿ

$$= (1)^2 + (2)^2 + (6)^3 + (24)^4 + \dots$$

$$= 1 + 4 + 216 + 6 \quad \text{only last digit}$$

$$= 1 + 4 + 6 + 6$$

$$= 17$$

Ans = 7.

⑥ Find unit digit : (It is solve for unit digit)

$$(23)^{53!} \times (77)^{23!} \times (38)^{23!} \times (25)^{11!}$$

Solⁿ

$$= (3)^{53!} \times (7)^{23!} \times (8)^{23!} \times (5)^{11!}$$

$$\Rightarrow \begin{array}{r|l} 4 \overline{) 93} (23 & 4 \overline{) 23} (5 \\ \underline{8} & \underline{20} \\ 13 & 03 \\ \underline{12} & \\ 04 & \end{array} \quad \begin{array}{r|l} 4 \overline{) 23} (5 & 4 \overline{) 23} (8 \\ \underline{20} & \underline{20} \\ 03 & 03 \end{array}$$

= 4! above all $\div$ by 4

$$= (3)^4 \times (7)^4 \times (8)^4 \times (5)^4$$

$$= \underset{\text{last digit}}{1} \times 1 \times 6 \times 5 = 30$$

Ans = 0

factorial continue

⑦ Find unit digit :

$$9^{(9000!+9)} + 8^{(8000!+8)} + \dots + 2^{(2000!+2)}$$

Solⁿ

$$= 9^{(4+9)} + 8^{(4+8)} + \dots + 2^{(4+2)}$$

$$= 9^{13} + 8^{12} + 7^{11} + 6^{10} + 5^9 + 4^8 + 3^7 + 2^6$$

using methods

$$= 9 + (8)^4 + (7)^3 + (6)^2 + 5 + \cancel{6^4} + 3^3 + 2^2$$

$$= 9 + 6 + 3 + \cancel{6} + 5 + \cancel{6} + 7 + 4 = 46$$

Ans = 6

⑧ Can a factorial be defined for negative number?

Ans : - No

⑨ Find the perfect square among the following.

(op² attached)

a) $37! \times 38!$

b) $36! \times 35!$

c) $39! \times 40!$

d) $34! \times 37!$

$$a) 37! \times 38!$$

w.k.t



$$\boxed{n! = n \times (n-1)!}$$

$$37! \times 38(37!)$$

but -

$$(37!)^2 \times 38 \rightarrow \text{Not Sq.}$$

$$b) 36! \times 35!$$

$$= 36 \times (35!) \times (35!)$$

$$= 6^2 \times (35!)^2$$